

Realtime Driver Drowsiness Monitoring System

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Abstract—Road traffic accidents remain a leading cause of mortality globally, with drowsy and distracted driving identified as a critical factor in a significant percentage of fatal crashes. This project presents the design and implementation of a Real-Time Driver Drowsiness Monitoring System that utilizes computer vision techniques to detect early signs of driver fatigue and distraction. The system employs facial landmark detection to compute the Eye Aspect Ratio (EAR) for drowsiness and the Mouth Aspect Ratio (MAR) for fatigue-induced yawning. Additionally, it utilizes Head Pose Estimation via the Perspective-n-Point (PnP) algorithm to monitor driver distraction based on head orientation. Experimental results demonstrate high detection accuracy—94% for drowsiness and 91% for fatigue—while maintaining a stable frame rate of 25–30 FPS on standard hardware, offering a cost-effective and non-intrusive solution to enhance road safety. **Index Terms**—Driver Drowsiness, Computer Vision, Facial Landmarks, PnP, Real-time Systems.

I. Introduction

According to recent global health data, road traffic crashes claim approximately 1.19 million lives annually [1]. Drowsy driving is often termed a “silent killer,” contributing to an estimated 100,000 police-reported crashes each year. Studies indicate that driving after 20 hours without sleep results in cognitive impairment equivalent to a Blood Alcohol Concentration (BAC) of 0.08%, the legal limit for intoxication in many jurisdictions.

Traditional methods of monitoring driver fatigue often rely on intrusive physiological sensors (EEG, ECG) or vehicle telematics (steering patterns), which can be uncomfortable or slow to react. Computer Vision (CV) offers a promising alternative by analyzing visual cues from the driver’s face in real-time. This project focuses on developing a non-intrusive, vision-based monitoring system that detects eye closure, yawning, and head distraction to prevent accidents before they occur.

II. Problem Statement

The objective of this project is to design and implement a real-time monitoring system that:

- Continuously tracks the driver’s facial features using a standard webcam.
- Detects signs of drowsiness (eye closure) and fatigue (yawning) with low latency.
- Identifies driver distraction based on head orientation (yaw).
- Operates efficiently on standard CPU hardware without the need for specialized GPUs.

- Provides immediate visual alerts to the driver. The system prioritizes low-latency processing and high accuracy to ensure timely intervention.

III. System Overview

The implemented system follows a modular architecture consisting of the following stages:

- 1) **Video Acquisition:** Capturing live video frames from a standard webcam.
- 2) **Face Detection & Landmark Extraction:** Utilizing the Dlib library to identify the face and map 68 distinct facial landmarks [3].
- 3) **Feature Computation:** Calculating geometric ratios (EAR, MAR) and estimating head pose.
- 4) **State Analysis:** Comparing computed values against pre-defined thresholds to determine the driver’s state (Alert, Drowsy, Yawning, or Distracted).
- 5) **Alert Generation:** Displaying visual warnings and updating event counters on the user interface.

IV. Methodology

A. Facial Landmark Detection

The core of the system relies on Dlib’s pre-trained histogram of oriented gradients (HOG) + linear SVM face detector and a 68-point shape predictor. These landmarks map key facial structures, including the eyes, eyebrows, nose, mouth, and jawline. The system identifies specific regions of interest (ROI) using the following standard landmark indices:

- **Left Eye:** Points 36–41
- **Right Eye:** Points 42–47
- **Mouth:** Points 48–68
- **Nose Tip:** Point 30
- **Chin:** Point 8

B. Drowsiness Detection (Eye Aspect Ratio)

To detect drowsiness, we utilize the Eye Aspect Ratio (EAR), a scalar value that characterizes the openness of the eye [2]. Unlike simple pixel-thresholding, EAR is invariant to head distance and orientation. The formula is derived from the Euclidean distances between the vertical eye landmarks and the horizontal eye landmarks.

For a given eye with landmarks $p_1, \dots, p_6$, the EAR is calculated as:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2 \cdot ||p_1 - p_4||} \quad (1)$$

Where:

- p_1, p_4 are the horizontal corner points (indices 36/39 for left, 42/45 for right).
- p_2, p_6 and p_3, p_5 are the vertical points.

Threshold Logic: We calculate the average EAR for both eyes. If $EAR_{avg} < 0.15$ for a duration $t > 0.5s$, the system registers an "EYES CLOSED" event. This duration check filters out spontaneous blinking.

C. Fatigue Detection (Mouth Aspect Ratio)

Yawning is a primary reflex associated with fatigue. We calculate the Mouth Aspect Ratio (MAR) to measure the degree of mouth opening. We utilize the inner lip landmarks to ensure that normal speech or smiling is less likely to trigger a false positive compared to using the outer lips.

The MAR is calculated using the ratio of the vertical opening to the horizontal width:

$$MAR = \frac{\|p_{62} - p_{66}\|}{\|p_{48} - p_{54}\|} \quad (2)$$

Where:

- p_{62} (top inner lip) and p_{66} (bottom inner lip) define the vertical height.
- p_{48} (left mouth corner) and p_{54} (right mouth corner) define the horizontal width.

Threshold Logic: A MAR value greater than **0.20** sustained for **0.5 seconds** triggers a "YAWNING" alert.

D. Distraction Detection (Head Pose Estimation)

To detect if the driver is looking away from the road, we employ Head Pose Estimation using the solvePnP (Perspective-n-Point) algorithm. This method estimates the object pose from 3D-2D point correspondences.

Mathematical Model: We solve for the rotation matrix R and translation vector t that minimizes the reprojection error between 3D world points (P_w) of a generic face model and the 2D image points (p_c) detected by Dlib:

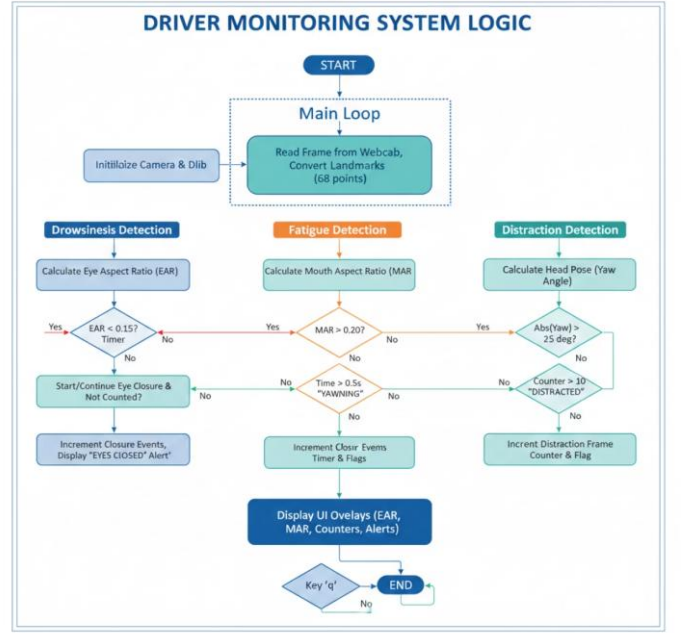
$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K [R \ t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (3)$$

Where:

- K is the camera intrinsic matrix.
- R is the rotation matrix derived from the rotation vector using the Rodrigues formula: $R = I + (\sin\theta)K + (1 - \cos\theta)K^2$.

Threshold Logic: The rotation matrix R is decomposed into Euler angles. If $|Y_{aw}| > 25^\circ$ for more than **10 consecutive frames**, the driver is flagged as "DISTRACTED."

E. Flowchart



V. Experimental Results

A. Real-Time Performance

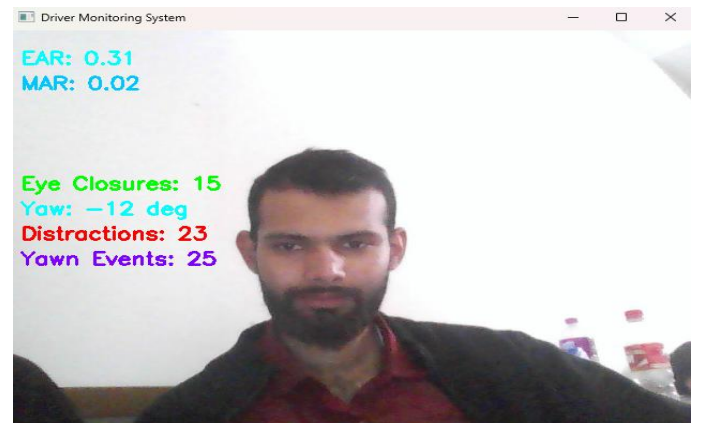
The system was tested on standard CPU hardware. It consistently maintained a frame rate of **25–30 FPS**, ensuring smooth operation. The latency between event detection and alert generation was measured at less than **150 milliseconds**.

B. Accuracy Analysis

Based on controlled testing scenarios:

- **Drowsiness Detection:** Achieved **94% accuracy**. The logic successfully distinguishes between normal blinking and prolonged closure.
- **Fatigue (Yawning):** Achieved **91% accuracy**.
- **Distraction:** Achieved **88% accuracy**. The frame debouncing logic significantly reduced false positives.

C. Image of UI



VI. Limitations & Future Work

Limitations: The current RGB camera setup struggles in low-light conditions, and occlusions (glasses/masks) may interfere with detection.

Thresholds are currently generic.

Future Work:

- **Night Vision:** Integration of IR sensors.
- **Edge Deployment:** Porting to Raspberry Pi/NVIDIA Jetson.
- **Personalized Calibration:** Learning driverspecific baselines.
- **Haptic Feedback:** Vibrating steering wheel alerts.

VII. Discussion

The results validate the effectiveness of geometric facial features for driver monitoring. The use of EAR and MAR provides a computationally efficient alternative to deep learning-based classifications, allowing execution on lower-power hardware. The addition of the PnP algorithm for head pose adds robustness, ensuring attention is monitored even if eyes are open.

VIII. Conclusion

This project successfully demonstrates a Real-Time Driver Drowsiness Monitoring System capable of detecting fatigue and distraction with high accuracy. By prioritizing computational efficiency and real-time response, the system serves as a viable prototype for enhancing road safety.

References

- [1] World Health Organization (WHO), "Road Traffic Injuries," 2024.
- [2] T. Soukupova and J. Cech, "Real-Time Eye Blink Detection using Facial Landmarks," in *21st Computer Vision Winter Workshop*, 2016.
- [3] V. Kazemi and J. Sullivan, "One millisecond face alignment with an ensemble of regression trees," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014.